

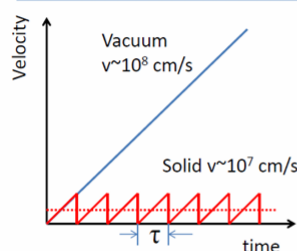
## Review

1. Midterm all up to include MOSFET
2. 5-6 problems
3. No sheet but provide all constants
4. Bring calculator no phone

## Lecture 1

## Lecture 2

### Electron in Motion



Electron's average velocity proportional to field

$$v = -\mu E$$

$\mu$ : Mobility

$$[\mu] = \frac{[v]}{[E]} = \frac{a\tau}{[E]} = \frac{q\tau}{m}$$

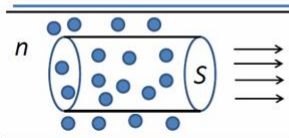
$$[\mu] \sim \frac{\text{cm/s}}{\text{V/cm}} = \frac{\text{cm}^2}{\text{V} \cdot \text{s}}$$

Example:

$$|E| = 10^4 \text{ V/cm}, \mu = 200 \text{ cm}^2/(\text{V} \cdot \text{s})$$

$$|v| = \mu |E| = 2 \times 10^6 \text{ cm/s}$$

### Current density



Imagine a group of electrons are passing through a wire with speed  $v$ . Pick a small area  $S$  perpendicular to the flow direction.

During a short time period  $t$ :

- Which electrons can pass through area  $S$ ? All the electrons in the cylinder with area  $S$  and length of  $vt$
- How many of them are there?  $N_e = n \cdot V = n \cdot vt \cdot S$
- How many charges do they carry?  $Q = (-q)N_e = -qnv t S$
- What is the current?  $I = Q/t = -qnv S$
- What is the current density?  $j = I/S = -qnv$

### Current Density & Conductivity

Current density = product of **carrier charge**, **carrier density**, and **velocity**  
( = product of **charge density** and **velocity** )

$$j = -q n v \rightarrow \text{Electron velocity [cm/s]}$$

Charge on each Electron [C]  $\leftarrow$   $\rightarrow$  Electron density [ $\text{cm}^{-3}$ ]

Intrinsic Ohm's Law: (**Drift** current density in an electric field)

$$j = -qnv = -qn(-\mu E) = qn\mu E = \sigma E$$

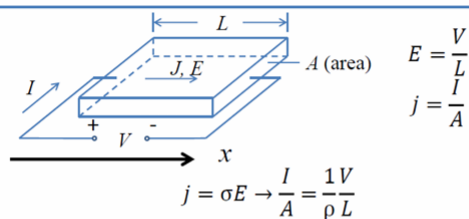
where  $\sigma = qn\mu$  is the **conductivity**.

$$[\sigma] = \frac{[j]}{[E]} = \frac{\text{A/cm}^2}{\text{V/cm}} = \frac{1}{\Omega \cdot \text{cm}}$$

$$\text{Resistivity: } \rho = \frac{1}{\sigma} [\Omega \cdot \text{cm}]$$



## Resistance & Conductance



Extrinsic Ohm's Law

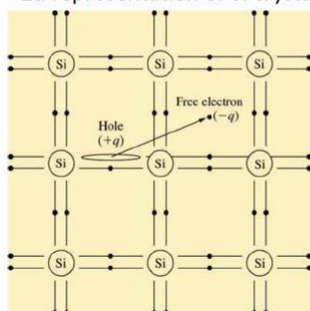
$$V = \rho \frac{L}{A} I = RI$$

where  $R = \rho \frac{L}{A}$  is the **resistance** [ $\Omega$ ].

**Conductance:**  $G = \frac{1}{R} = \sigma \frac{A}{L}$  [ $\Omega^{-1}$ ].

## Carrier Concentration - Silicon

2d representation of Si crystal structure ( $T > 0$  K)



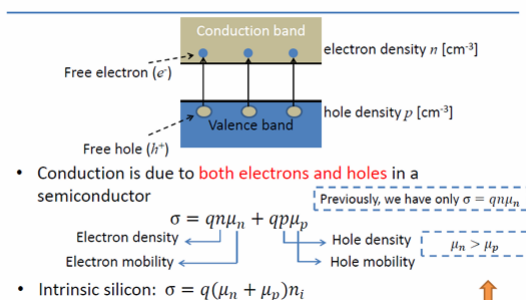
- At  $T > 0$  K, thermal energy inside the crystal can excite small amount of bound electrons into **free electrons**, leaving a **hole** in the bond
- Density of these free electrons (and also holes) is called the **intrinsic carrier concentration**

**Intrinsic semiconductor** 指的是纯净半导体，没有人掺杂 donor 或 acceptor。

与之相对的是：extrinsic semiconductor

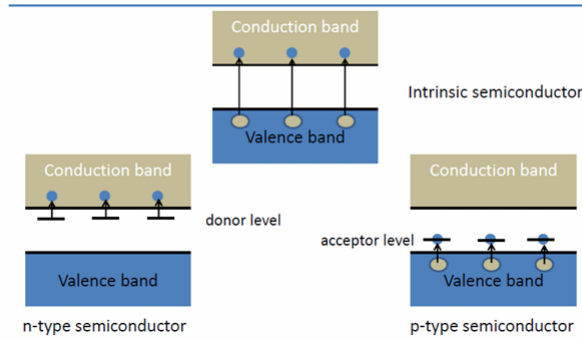
对温度极其敏感。室温下硅常用  $10^{10} \text{ cm}^{-3}$ ，但换温度就不能随便用了。

## Conduction in Intrinsic Silicon



电子只能出现在导带或者价带，不能出现在禁带区域中

## Conduction in Intrinsic Silicon



P type 和 N type 都会提高电导率

## Extrinsic Semiconductors

### – Carrier Density

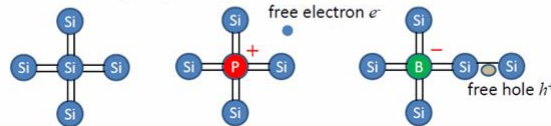
- The carrier density ( $n, p$ ) in extrinsic semiconductors is governed by two laws:

– Law of Mass Action (equilibrium condition):

$$np = n_i^2$$

– Charge Neutrality Condition:

$$N_D + p - N_A - n = 0$$



没有外部输入的静态的就是热平衡状态

### Extrinsic Semiconductors

#### – Carrier Density ( $n$ -type)

- Silicon doped with donors with density  $N_D$  ( $N_A = 0$ )
  - Solve  $np = n_i^2$  and  $N_D + p - n = 0$
  - Eliminate  $p$ :
 
$$N_D + \frac{n_i^2}{n} - n = 0 \Rightarrow n^2 - nN_D - n_i^2 = 0$$
  - Solve:  $n = \frac{N_D + \sqrt{N_D^2 + 4n_i^2}}{2}$ ,  $p = \frac{n_i^2}{n}$
  - Most of the time:  $N_D \gg n_i$ , thus  $n = N_D$ ,  $p = \frac{n_i^2}{N_D} \Rightarrow n \gg n_i \gg p$ , the semiconductor is called " $n$ -type"
  - Electrons: **majority** carrier; Holes: **minority** carrier

### Extrinsic Semiconductors

#### – Carrier Density ( $p$ -type)

- Silicon doped with acceptors with density  $N_A$  ( $N_D = 0$ )
  - Solve  $np = n_i^2$  and  $p - N_A - n = 0$
  - Eliminate  $n$ :
 
$$p - N_A - \frac{n_i^2}{p} = 0 \Rightarrow p^2 - pN_A - n_i^2 = 0$$
  - Solve:  $p = \frac{N_A + \sqrt{N_A^2 + 4n_i^2}}{2}$ ,  $n = \frac{n_i^2}{p}$
  - Most of the time:  $N_A \gg n_i$ , thus  $p = N_A$ ,  $n = \frac{n_i^2}{N_A} \Rightarrow p \gg n_i \gg n$ , the semiconductor is called " $p$ -type"
  - Holes: **majority** carrier; Electrons: **minority** carrier

### 2.6.1 N 型材料 ( $N_D > N_A$ )

由式(2.2)解出  $p$ , 代入式(2.10)得到关于  $n$  的二次方程

$$n^2 - (N_D - N_A)n - n_i^2 = 0$$

则

$$n = \frac{(N_D - N_A) + \sqrt{(N_D - N_A)^2 + 4n_i^2}}{2} \quad \text{和} \quad p = \frac{n_i^2}{n} \quad (2.11)$$

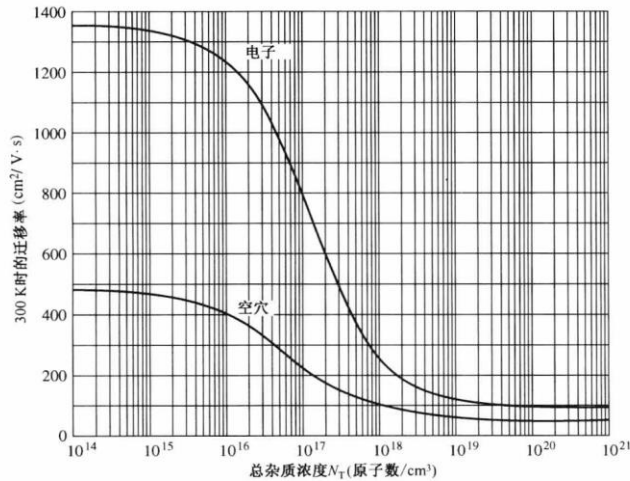
上式适用于  $N_D > N_A$  的情况。实际掺杂中,  $N_D - N_A \gg 2n_i$ , 因此  $n \approx (N_D - N_A)$ 。

### 2.6.2 P 型材料 ( $N_A > N_D$ )

$N_A > N_D$  时, 由式(2.2)解出  $n$ , 代入式(2.10)得到关于  $p$  的二次方程

$$p = \frac{(N_A - N_D) + \sqrt{(N_A - N_D)^2 + 4n_i^2}}{2} \quad \text{和} \quad n = \frac{n_i^2}{p} \quad (2.12)$$

上式适用于  $N_A > N_D$  的情况。一般  $N_A - N_D \gg 2^2 n_i$ ,  $p \approx (N_A - N_D)$ 。

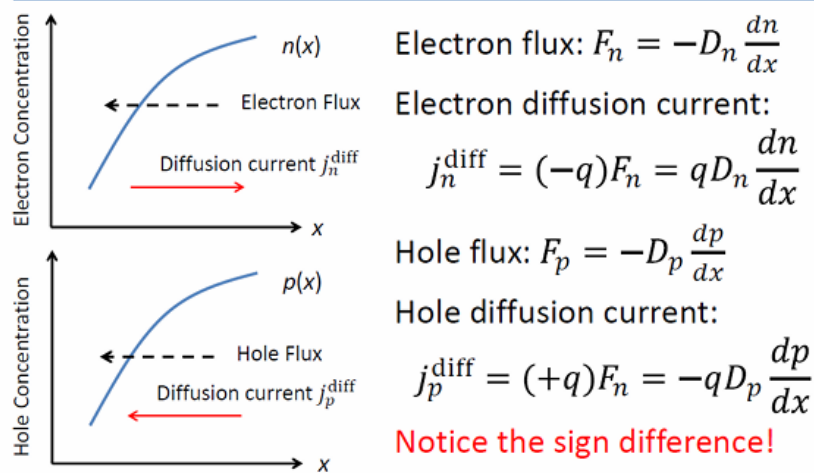


迁移率近似表达式

$$\mu_n = 92 + \frac{1270}{1 + \left( \frac{N_T}{1.3 \times 10^{17}} \right)^{0.91}}$$

$$\mu_p = 48 + \frac{447}{1 + \left( \frac{N_T}{6.3 \times 10^{16}} \right)^{0.76}}$$

## Conduction Mechanisms – Diffusion Current



## Charge Transport in Semiconductors

- Total current = drift + diffusion

$$j_n^{\text{tot}} = j_n^{\text{drift}} + j_n^{\text{diff}} = qn\mu_n E + qD_n \frac{dn}{dx}$$

$$j_p^{\text{tot}} = j_p^{\text{drift}} + j_p^{\text{diff}} = qp\mu_p E - qD_p \frac{dp}{dx}$$

- 3-dimensional case:

$$\mathbf{j}_n^{\text{tot}} = qn\mu_n \mathbf{E} + qD_n \nabla n$$

$$\mathbf{j}_p^{\text{tot}} = qp\mu_p \mathbf{E} - qD_p \nabla p$$

# Einstein Relationship

- $D_n, D_p$ : diffusivity of electrons and holes
- $\mu_n, \mu_p$ : mobility of electrons and holes
- Einstein's relationship:

$$\frac{D}{\mu} = \frac{kT}{q}$$

Rhymes both forwards and backwards!

- $kT/q$  is called “**thermal voltage**” ( $V_T$ )
- At 300 K,  $\frac{kT}{q} = \frac{8.62 \times 10^{-5} \text{ eV/K} \times 300 \text{ K}}{1e} = \mathbf{25.9 \text{ mV}}$

k 是一个常数

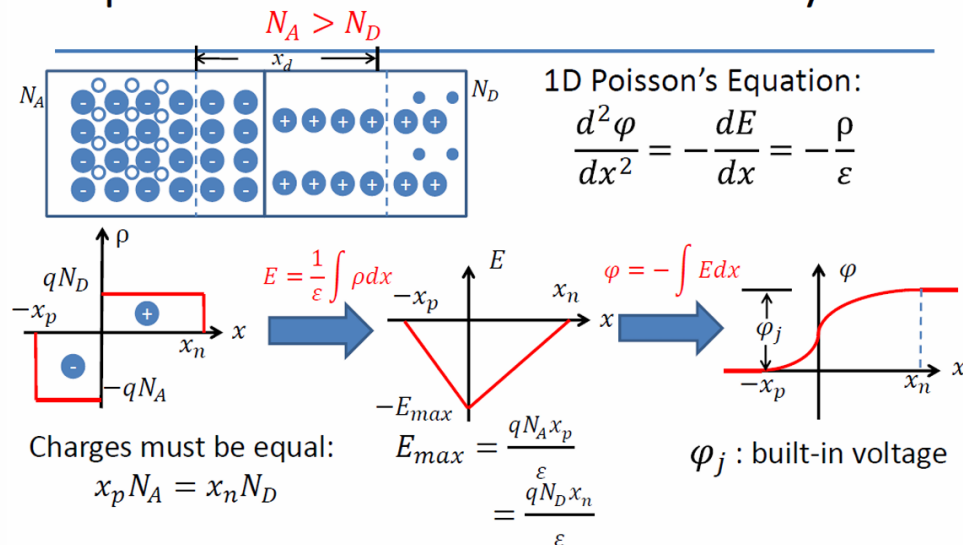
pn diode 规定:

p-side = anode

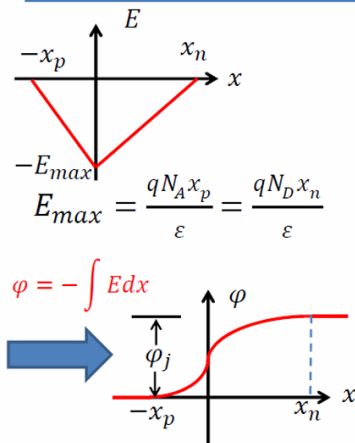
n-side = cathode

Lecture 3

## p-n Junction – Electrostatic Analysis



## p-n Junction – Electrostatic Analysis



- $\phi_j$  is the area under E-x curve:
  - $\phi_j = \frac{(x_n + x_p)}{2} |E_{max}| = \frac{qN_A(x_n + x_p)}{2\epsilon} x_p$
  - Combining  $x_p N_A = x_n N_D$ , we get
- $$x_p = \sqrt{\frac{2\epsilon\phi_j}{qN_A(1+N_A/N_D)}}$$
- $$x_n = \sqrt{\frac{2\epsilon\phi_j}{qN_D(1+N_D/N_A)}}$$
- $$x_d = x_n + x_p = \sqrt{\frac{2\epsilon\phi_j}{q} \left( \frac{1}{N_D} + \frac{1}{N_A} \right)}$$

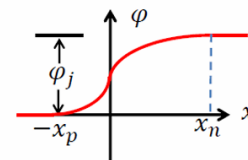
Depletion width under zero bias

## p-n Junction – Electrostatic Analysis

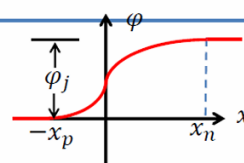
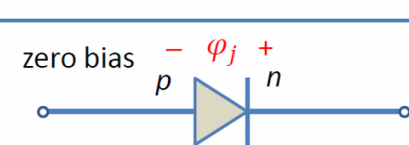
- How to find  $\phi_j$  ?
- From detailed balance in equilibrium ( $J^{\text{drift}} = -J^{\text{diff}}$ ):

$$\phi_j = \frac{kT}{q} \ln \frac{p_0(-x_p)}{p_0(x_n)} = \frac{kT}{q} \ln \frac{n_0(x_n)}{n_0(-x_p)} \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$

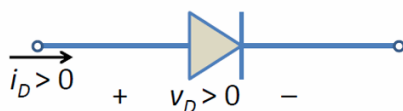
- $\phi_j$  depends on
  - Doping concentration on both sides ( $N_A, N_D$ )  $= V_T \ln \frac{N_A N_D}{n_i^2}$
  - Temperature ( $kT$ )
  - Material ( $n_i^2 \sim \exp[-E_g/kT]$ )



## p-n Junction Under Applied Voltage



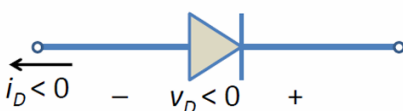
forward bias



- Under bias  $v_D$ , replace  $\phi_j$  with  $(\phi_j - v_D)$ . e.g.

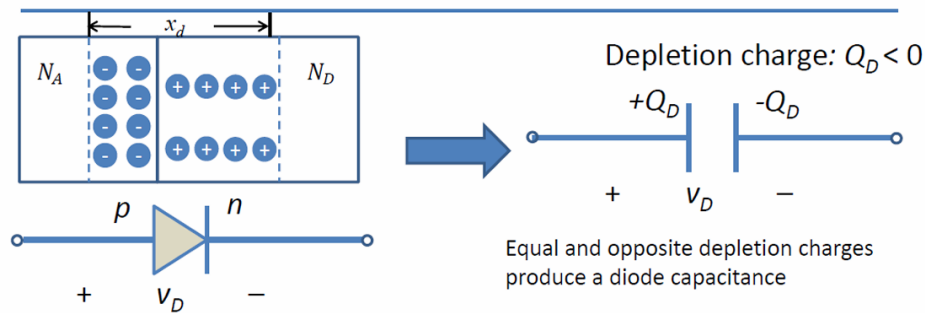
$$x_d = \sqrt{\frac{2\epsilon(\phi_j - v_D)}{q} \left( \frac{1}{N_D} + \frac{1}{N_A} \right)}$$

reverse bias



- Forward bias,  $v_D > 0$ ,  $x_d$  shrinks
- Reverse bias,  $v_D < 0$ ,  $x_d$  widens

## p-n junction capacitance



- Normal linear capacitance (e.g. parallel plate):

$$C = \frac{Q}{V}$$

- "Depletion capacitance" or "junction capacitance" for diodes:

$$C_j = \frac{dQ_D}{dv_D}$$

- Depletion capacitance:

$$Q_D = -qAx_pN_A$$

$$x_p = \sqrt{\frac{2\epsilon_s(\phi_j - v_D)}{qN_A(1 + N_A/N_D)}}$$



$$C_j = \frac{dQ_D}{dv_D} = \frac{\epsilon_s A}{x_d}$$

A: diode area [cm<sup>2</sup>]

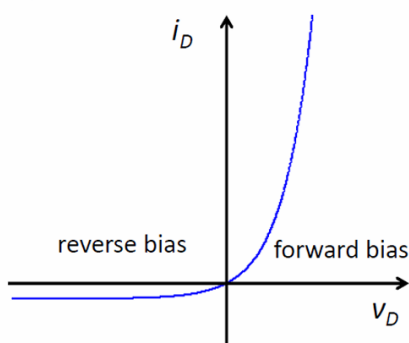
- Looks like a parallel-plate capacitor formula!
- As  $x_d$  varies with  $v_D$ ,  $C_j$  also varies with  $v_D$ .

– At zero bias,  $C_{j0} = \frac{\epsilon_s A}{x_{d0}}$

Voltage-dependent capacitor: "varactor"

– At a non zero bias,  $C_j = \frac{\epsilon_s A}{x_d} = C_{j0} \frac{x_{d0}}{x_d} = C_{j0} \sqrt{\frac{\phi_j}{\phi_j - v_D}}$

## Diode I-V Characteristics



- Shockley diode equation:

$$i_D = I_S(e^{\frac{v_D}{nV_T}} - 1)$$

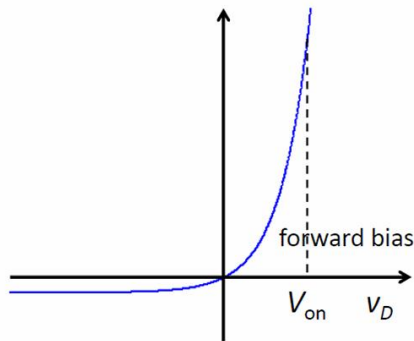
- $I_S$ : Reverse saturation current typically,  $10^{-18} \text{ A} \leq I_S \leq 10^{-9} \text{ A}$  (small!)
- $n$ : nonideality factor,  $n=1$  for ideal diodes (Assume  $n=1$  later on)
- $V_T$ : thermal voltage
- Can solve for  $v_d$ :

$$v_D = nV_T \ln\left(\frac{i_D}{I_S} + 1\right)$$

Let  $I_S = 10^{-13} \text{ A}$ ,  $n=1$ . Then if  $i_D = 1 \text{ mA}$ ,  $v_d = 0.026 \ln(10^{10} + 1) = 0.60 \text{ V}$

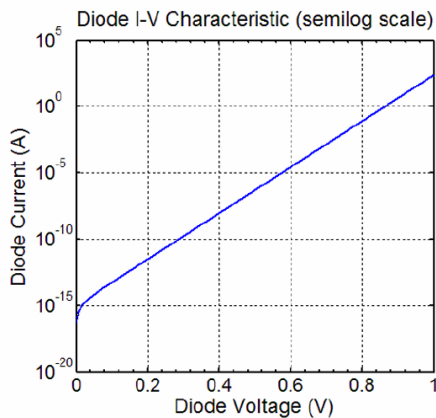


# Diode I-V Forward Bias



- Shockley diode equation:  

$$i_D = I_S(e^{\frac{v_D}{V_T}} - 1)$$
- Forward bias,  $i_D$  large.
- If  $v_D \gg 4V_T$ , then  $e^{\frac{v_D}{V_T}} \gg 1$   
 $\Rightarrow i_D = I_S e^{\frac{v_D}{V_T}}$  (exponential)
- Turn-on voltage:** Voltage required to bring the diode into significant conduction (typically  $\sim 0.7$  V).



- Exponential increase  

$$i_D = I_S e^{\frac{v_D}{V_T}}$$
- $\Delta V_{\text{decade}}$ : Voltage increase required to increase current by a factor of 10.
  - $\Delta V_{\text{decade}} = V_T \ln 10$
  - At RT,  $\Delta V_{\text{decade}} = 60 \text{ mV} \Rightarrow$   
 A 60 mV increase in  $v_D$  gives a **decade** increase in  $i_D$   
 (60n mV per decade if nonideal)

## Diode reverse leakage current

Short base  $W_p \ll L_n, W_n \ll L_p$ :

$$I_S = qAn_i^2 \left[ \frac{D_n}{N_A W_p} + \frac{D_p}{N_D W_n} \right]$$

Long base  $W_p \gg L_n, W_n \gg L_p$ :

$$I_S = qAn_i^2 \left[ \frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right]$$

Current dominated by minority carrier injection into the lightly-doped and/or narrower side.

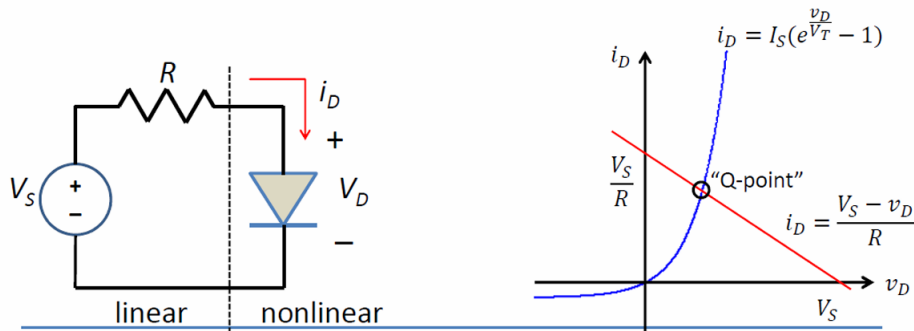


## Load Line Analysis (Graphical)

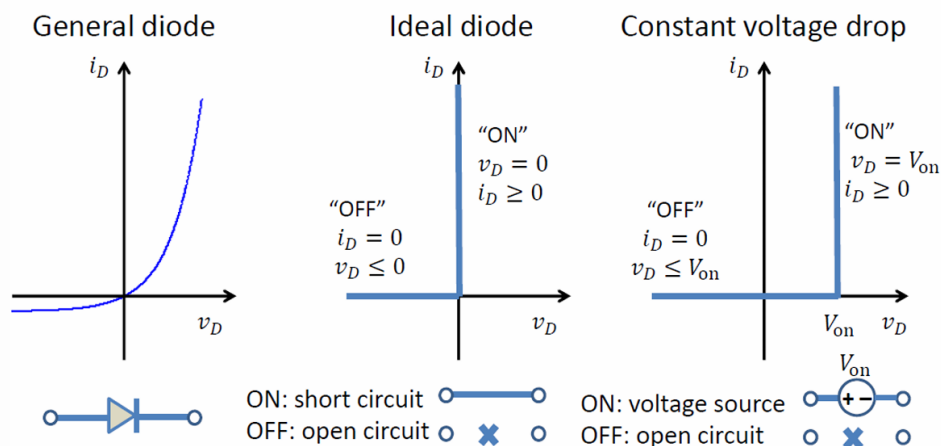
- Write the two equations in the form of  $i_D$  v.s.  $v_D$ .

$$i_D = -\frac{v_D}{R} + \frac{V_S}{R}, \quad i_D = I_S(e^{\frac{v_D}{V_T}} - 1)$$

- Plot them on a same graph, find the intersection.



## Simplified Diode Models



Lecture 4

## EXAMPLE

### Two-Diode Circuit Analysis

#### 5 steps to solve diode circuits

- Select a diode model
- Label  $i_D$  and  $v_D$
- Guess each diode's region of operation from circuit.
- Analyze circuit using the equivalent circuit solve the currents and voltages at any node.
- Verify that the equivalent replacements were valid. If not, guess another state for the diode.

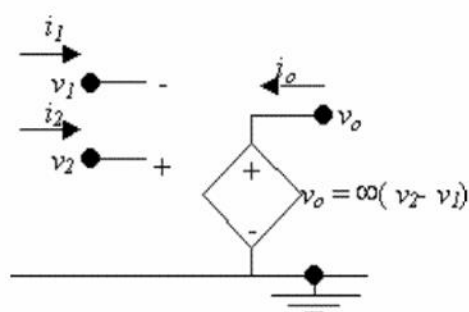
只要你试出一种状态，并且所有二极管的验证条件都满足，就可以停止，不需要继续尝试其他组合。

| Diode model                       | 导通条件                          | ON 等效                            | 截止条件                    | OFF 等效 |
|-----------------------------------|-------------------------------|----------------------------------|-------------------------|--------|
| Ideal diode model 理想模型            | $v_D = 0, i_D \geq 0$         | 短路, $v_D = 0$                    | $v_D \leq 0, i_D = 0$   | 开路     |
| Constant voltage drop model 恒压降模型 | $v_D = V_{ON}, i_D \geq 0$    | 电压源, 硅管常取 $v_D = 0.7V$           | $v_D < V_{ON}, i_D = 0$ | 开路     |
| Piecewise-linear model 分段线性模型     | $v_D \geq V_{ON}, i_D \geq 0$ | $v_D = V_{ON} + i_D r_D$         | $v_D < V_{ON}, i_D = 0$ | 开路     |
| Exponential model 指数模型            | 没有硬阈值; $v_D > 0$ 时电流指数增大      | $i_D = I_S (e^{v_D/(nV_T)} - 1)$ | $v_D < 0$ 时只有很小反向电流     | 同一指数公式 |

## Lecture 5

### Ideal op amp relationships

An ideal op amp is an ideal voltage controlled voltage source. We can think of the op amp symbol being replaced as follows:



Notice what this implies: If  $v_o$  is finite, then

$$v_2 - v_1 = \frac{v_o}{\infty} = 0$$

$$v_1 = v_2$$

and

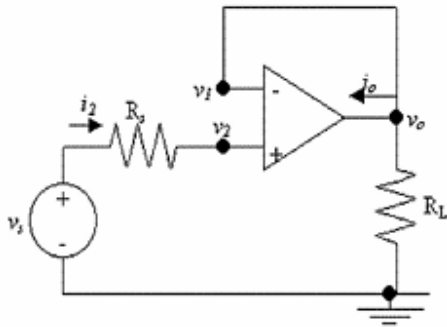
$$i_1 = 0$$

$$i_2 = 0$$

These are, in fact, the *ideal op amp relationships*, and once we get used to them we can use them to solve op amp circuits.

## Op amp circuit analysis

Let's try a circuit with an op amp.  
What's the relationship between  $v_s$  and  $v_o$ ?



$v_o = v_1$  since they are connected.

$v_o = v_1 = v_2$  from the ideal op amp relationships.

Remember that there is no current flowing into the op amp inputs. So

$i_2 = 0$ ,  $v_2 = v_s$  and thus

$v_o = v_s$

See how the special op amp relationships simplify circuit analysis?

因为输出电压跟随输入电压：

$$v_o = v_s$$

电压增益：

$$A_v = \frac{v_o}{v_s} = 1$$

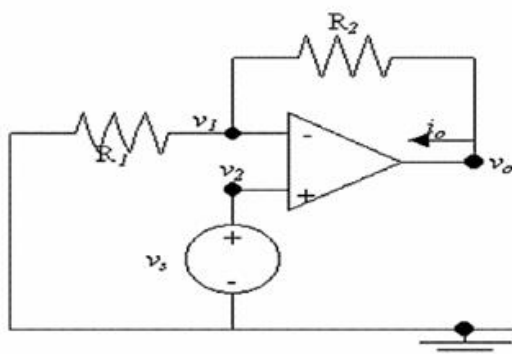
### 3. 如果没有 op amp, 会发生什么?

假设信号源有内阻  $R_s$ , 负载是  $R_L$ 。

如果直接连接, 那么输出到负载上的电压是分压：

$$v_L = v_s \frac{R_L}{R_s + R_L}$$

## More analysis-2



Now, write KCL at  $v_1$ :

$$\frac{v_s}{R_1} + \frac{v_s - v_o}{R_2} = 0$$

$$v_s \left( \frac{1}{R_1} + \frac{1}{R_2} \right) - \frac{v_o}{R_2} = 0$$

$$v_o = v_s \left( 1 + \frac{R_2}{R_1} \right)$$

#### Key term

• Potentiometer  
A three-terminal resistor with a center sliding contact (wiper) that forms a voltage divider or a variable resistor



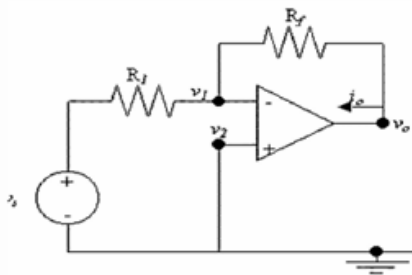
This is called a **non-inverting amplifier**. Notice that the gain of the amplifier can be changed by changing the ratio of the resistors, for example with a potentiometer for  $R_2$ .

## Design with op amps-2

Example: We want to amplify an input signal by a factor of 100 (gain of 100).  
The input signal has a range of 0 to -100 mV, but we want a positive voltage output.

Pretty clearly we want the inverting amplifier circuit from the catalog:

Please wait while I thumb through the catalog. OK, I found it:



Let's check the analysis:

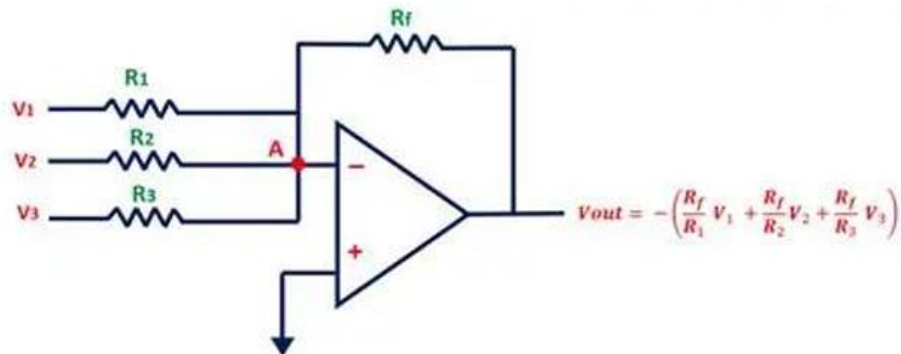
$$\begin{aligned} v_2 &= 0 \\ v_1 &= 0 \\ \frac{v_s - 0}{R_i} + \frac{v_o - 0}{R_f} &= 0 \\ v_o &= -\frac{R_f}{R_i} v_s \end{aligned}$$



Robert Widlar inventor of the first mass-produced op amp, uA709

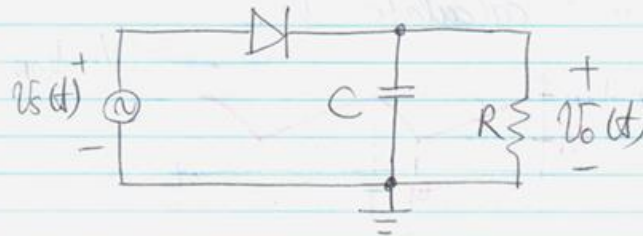
## Summer

A superposition of multiple inverter:

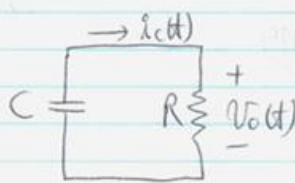


⑤ Half-wave rectifier with RC load.

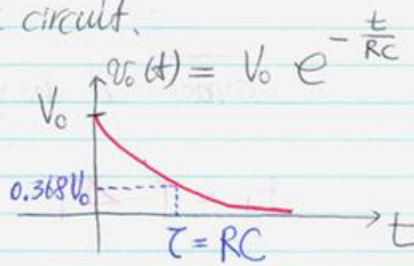
To make use of the output voltage, a load must be connected to the circuit as represented by the resistor " $R$ ."



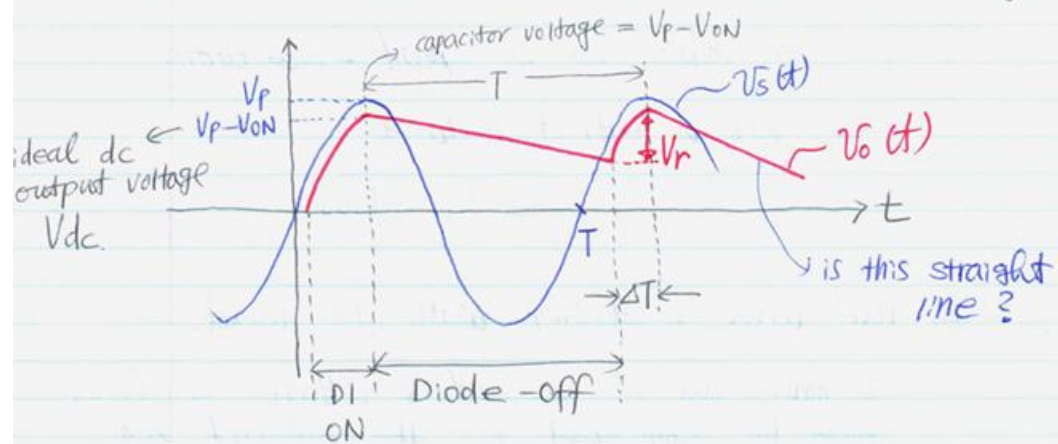
\* Source-free RC circuit.



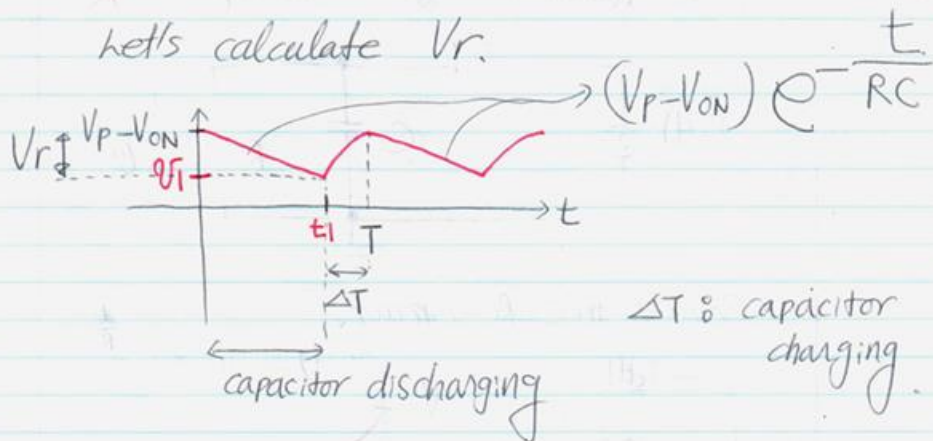
$$v_o(t=0) = V_0$$



→ larger value of  $\tau$  (time constant) provides slower dissipation of the stored energy.



let's calculate  $V_r$ .



$$t_1 = T - \Delta T$$

$$\begin{aligned}
 V_r &= (V_p - V_{on}) - V_1 \\
 &= (V_p - V_{on}) - (V_p - V_{on}) e^{-\frac{T - \Delta T}{RC}} \\
 &= (V_p - V_{on}) \left[ 1 - e^{-\frac{T - \Delta T}{RC}} \right]
 \end{aligned}$$

in order to minimize  $V_r$ ,  $RC \gg T - \Delta T$   
(slow dissipation)

Using  $e^{-x} \approx 1 - x$  for  $x \ll 1$

discharge waveform is a straight line.

$$\begin{aligned}
 V_r &\approx (V_p - V_{on}) \left[ 1 - \left( 1 - \frac{T - \Delta T}{RC} \right) \right] \\
 &\approx (V_p - V_{on}) \frac{T - \Delta T}{RC} \\
 &\approx (V_p - V_{on}) \frac{T}{RC} \left( 1 - \frac{\Delta T}{T} \right)
 \end{aligned}$$

A small ripple voltage means  $\Delta T \ll T$

$$\therefore V_r \approx \frac{V_p - V_{on}}{R} \frac{T}{C}$$



$$\therefore v_s(t_1) = V_p \sin \omega(t' + \frac{T}{4})$$

$$= V_p \cos \omega t'$$

$$= V_p \cos \omega(T - \Delta T)$$

$$= V_p \cos(\omega \Delta T)$$

$$\cos(\omega t) = \cos(-\omega t)$$

$$= \cos(T - \omega t)$$

$$\triangleq \theta_c \text{ (conduction angle)}$$

\* Taylor series :

$$\cos(x) \cong 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\text{when } |x| < 1 \quad \frac{2\pi}{T} \Delta T < 1$$

$$v_s(t_1) \cong V_p \left( 1 - \frac{\omega^2 \Delta T^2}{2} \right) \text{ for } \boxed{\omega \Delta T < 1}$$

from the equation ①,

$$V_r \cong \frac{V_p - V_{on}}{R} \frac{T}{C}$$

$$v_s(t_1) - V_{on} = (V_p - V_{on}) - V_r$$

$$V_p \left( 1 - \frac{(\omega \Delta T)^2}{2} \right) - V_{on} = (V_p - V_{on}) - (V_p - V_{on}) \frac{T}{RC}$$

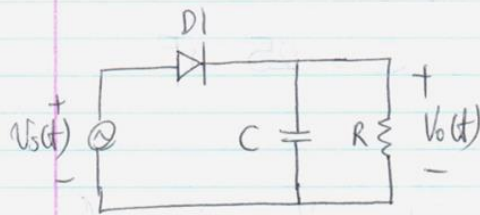
$$V_p \frac{(\omega \Delta T)^2}{2} = (V_p - V_{on}) \frac{T}{RC}$$

$$\Rightarrow \frac{(\omega \Delta T)^2}{2} = \frac{V_p - V_{on}}{V_p} \frac{T}{RC} = \frac{V_r}{V_p}$$

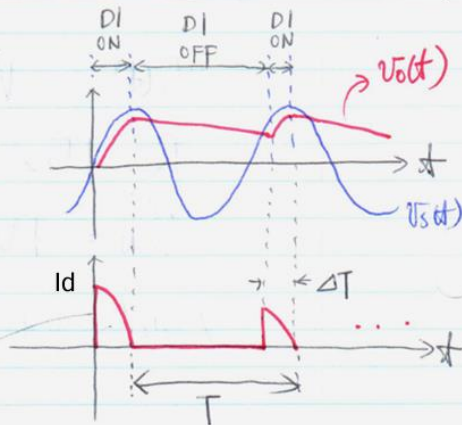
$$\Rightarrow \Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}} \text{ [sec]}$$

$$\theta_c = \omega \Delta T = \sqrt{2 \frac{V_r}{V_p}}$$

### ⑥ peak diode current

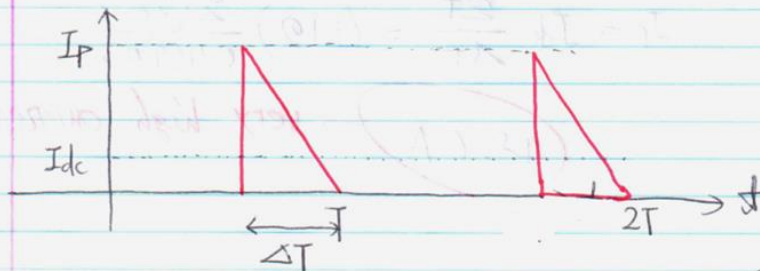


can be modeled as a triangle



- non-zero current exists in the diode for only a very small fraction of period  $T$ , yet an almost constant dc current flows out of the filter capacitor to load.  $I_{dc}$

- The total charge lost from the filter capacitor in each cycle is replenished by the diode during a short conduction interval causing high peak diode current.



$$Q = I_p \frac{\Delta T}{2} = I_{dc} T$$

←  $Q = \int i(t) dt$

change replenished by a diode
change lost from the capacitor.

## <Lecture 7>

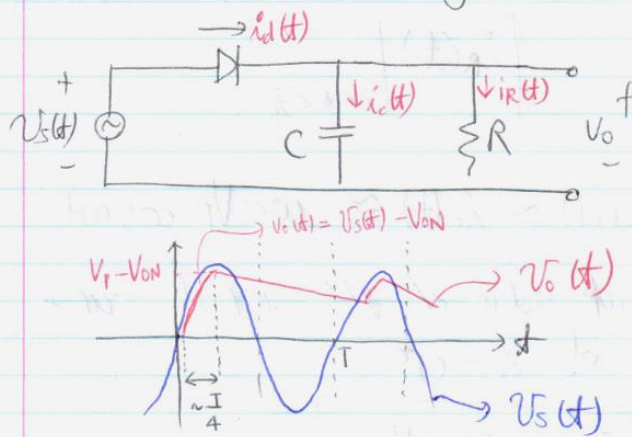
3.13.6

P 120

### ① Diode Surge Current ( $I_{sc}$ )

In addition to the peak diode currents, there is an even larger current through the diode called "**surge current**" that occurs when power is first turned on.

During first quarter cycle, current through diode is approximately,



$$i_d(t) = i_c(t) + i_R(t)$$

However,  $|i_c(t)| \gg |i_R(t)|$

$$|i_R(t)|_{\max} = \frac{V_P - V_{on}}{R}$$

$$i_c(t) = C \frac{dv(t)}{dt} = C \frac{d}{dt} (V_p \sin \omega t - V_{DN})$$

$$= C V_p \omega \cos(\omega t)$$

when  $t = 0^+$  (this means  $t \rightarrow 0$  as soon as the diode is turned on)

$$|i_c(t)| \approx C V_p \omega = C V_p 2\pi f$$

this value is typically larger than

$$|i_R(t)|_{0 < t < \frac{\pi}{4}}$$

$$\therefore i_d(t) \approx i_c(t) \approx \omega C V_p \cos \omega t$$

the peak value of this initial surge current occurs at  $t = 0^+$ :

$$I_{sc} = \omega C V_p$$

$$\text{if } C = 25 \text{ mF}, \quad V_p = 12.6\sqrt{2}$$

$$\text{then, } I_{sc} = (2\pi 60)(25 \text{ m})(12.6\sqrt{2})$$

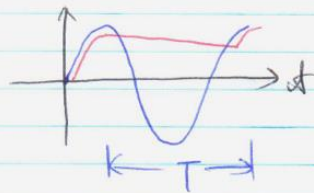
$$= 168 \text{ A}$$



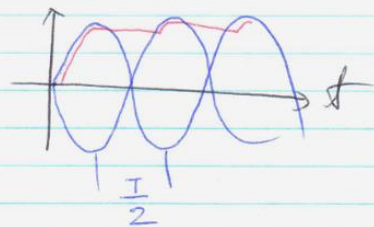
## ⑥ Rectifier Comparison.

(1) Assuming the same R & C

Half-wave



Full-wave



$$V_r = (V_p - V_{on}) \frac{T}{RC}$$

$$V_r = (V_p - V_{on}) \frac{T}{2RC}$$

$$\Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}} \text{ [sec]}$$

$$\Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}} \text{ [sec]}$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{2T}{\Delta T}$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{T}{\Delta T}$$

(2) Assuming the same ripple voltage ( $V_r$ )

Half-wave

Full-wave

Full-wave Bridge

$$C = \frac{V_p - V_{on}}{V_r} \frac{T}{R}$$

$$C = \frac{V_p - V_{on}}{V_r} \frac{T}{2R}$$

$$C = \frac{V_p - V_{on}}{V_r} \frac{T}{2R}$$

$$PIV = 2V_p$$

$$2V_p$$

$$V_p$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{2T}{\Delta T}$$

$$I_p = \frac{V_p - V_{on}}{R} \frac{T}{\Delta T}$$

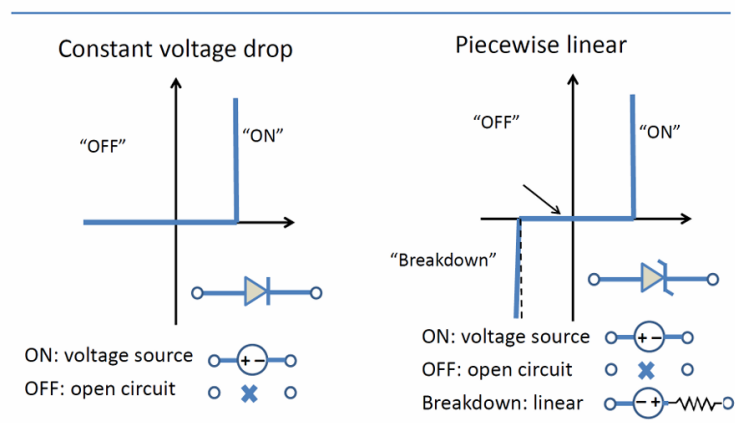
$$\frac{(V_p - 2V_{on}) * T}{(R * \Delta T)}$$

| Quantity                                 | Half-wave rectifier                                    | Full-wave rectifier                                    |
|------------------------------------------|--------------------------------------------------------|--------------------------------------------------------|
| Time between charging peaks $T_r$        | $T$                                                    | $\frac{T}{2}$                                          |
| Charging time / diode ON time $\Delta T$ | $\Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}}$ | $\Delta T = \frac{1}{\omega} \sqrt{2 \frac{V_r}{V_p}}$ |
| Discharging time $t_1$                   | $t_1 = T - \Delta T$                                   | $t_1 = \frac{T}{2} - \Delta T$                         |

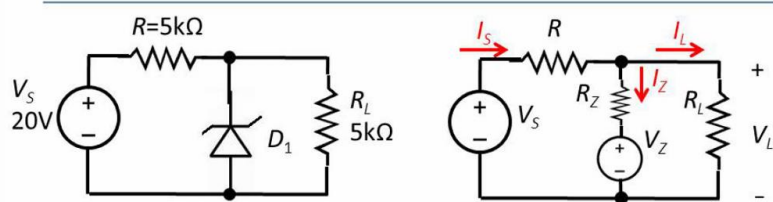
| Quantity                                        | Half-wave rectifier                                                 | Full-wave rectifier                                                    |
|-------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------|
| Ripple voltage $V_R$                            | $V_R \approx V_M(1 - e^{-(T-\Delta T)/(RC)})$                       | $V_R \approx V_M(1 - e^{-(T/2-\Delta T)/(RC)})$                        |
| Ripple voltage approximation                    | $V_R \approx \frac{V_M(T - \Delta T)}{RC} \approx \frac{V_M T}{RC}$ | $V_R \approx \frac{V_M(T/2 - \Delta T)}{RC} \approx \frac{V_M T}{2RC}$ |
| DC output voltage $V_{DC}$                      | $V_{DC} \approx V_M - \frac{V_R}{2}$                                | $V_{DC} \approx V_M - \frac{V_R}{2}$                                   |
| DC load current $I_{DC}$                        | $I_{DC} \approx \frac{V_{DC}}{R}$                                   | $I_{DC} \approx \frac{V_{DC}}{R}$                                      |
| Conduction angle $\theta_C$ under your notation | $\theta_C = \omega \Delta T \approx \sqrt{\frac{2V_R}{V_p}}$        | $\theta_C = \omega \Delta T \approx \sqrt{\frac{2V_R}{V_p}}$           |

## Lecture 7

### Piecewise linear model



### Voltage Regulation ( $R_Z \neq 0$ )

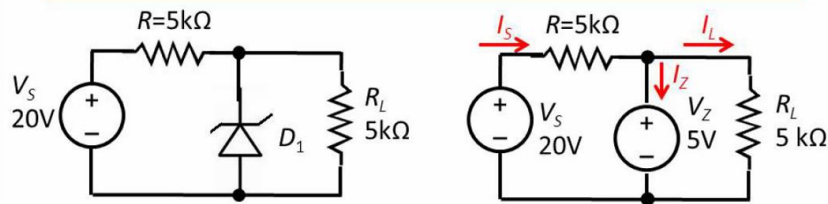


$$I_S = I_Z + I_L \Rightarrow \frac{V_S - V_L}{R} = \frac{V_L - V_Z}{R_Z} + \frac{V_L}{R_L}$$

$$V_L = \left( \frac{V_S}{R} + \frac{V_Z}{R_Z} \right) / \left( \frac{1}{R} + \frac{1}{R_L} + \frac{1}{R_Z} \right)$$

Minimum  $R_L$  unchanged ( $I_Z=0$  at threshold).

## Voltage Regulation ( $R_Z = 0$ )



Guess: Breakdown  $I_S = \frac{V_S - V_Z}{R} = 3 \text{ mA}$ ,  $I_L = \frac{V_Z}{R_L} = 1 \text{ mA}$   
 $I_Z = I_S - I_L = 2 \text{ mA} > 0$ .  
 $I_D = -I_Z < 0$ . Consistent with "breakdown"

$$I_Z = \frac{V_S - V_Z}{R} - \frac{V_Z}{R_L} > 0 \Rightarrow R_L > \frac{R}{\frac{V_S}{V_Z} - 1} = \frac{5 \text{ k}\Omega}{\frac{20 \text{ V}}{5 \text{ V}} - 1} = 1.67 \text{ k}\Omega$$

The minimum load resistance to maintain regulation is  $\frac{R}{\frac{V_S}{V_Z} - 1}$

## Voltage Regulation (constant $I_L$ )

- $V_L = \left( \frac{V_Z}{R_Z} + \frac{V_S}{R} - I_L \right) / \left( \frac{1}{R} + \frac{1}{R_Z} \right)$

- Line regulation

$$\frac{dV_L}{dV_S} = \frac{R_Z}{R + R_Z}$$

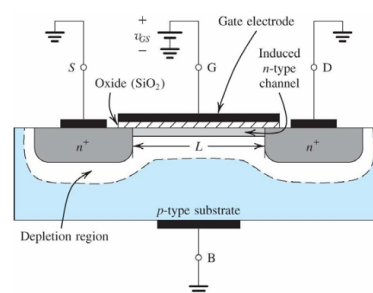
- Load regulation

$$\frac{dV_L}{dI_L} = -\frac{RR_Z}{R + R_Z}$$

### Lecture 8

## Creating a channel for current flow

- > With the drain and source terminals grounded,
- > A gate to source voltage,  $V_{GS}$  is applied to the gate terminal, causing a build up of positive charge along the metal electrode
- > This build up causes free holes to be repelled from region of p-type substrate
- > These holes are pushed to the substrate leaving behind a depletion region

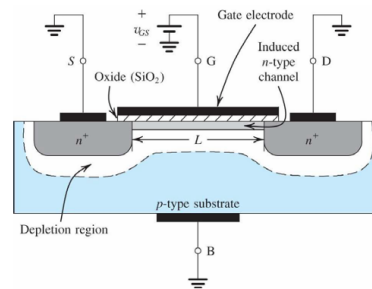


The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.



## Creating a channel for current flow

- › The positive gate voltage also attracts electrons from the n+ source and drain regions into the channel
- › Once a sufficient number of these electrons accumulate, an n-region is created, connecting the source and drain regions
- › This provides a path for current flow between Drain and Source



The enhancement-type NMOS transistor with a positive voltage applied to the gate. An n channel is induced at the top of the substrate beneath the gate.

## Creating a channel for current flow

- › The excess voltage over  $V_t$  is termed the overdrive voltage (in the textbook)

$$v_{OV} \equiv v_{GS} - V_t$$

- › The electron charge in the channel is expressed as

$$|Q| = C_{ox}(WL)v_{OV}$$

- › Where  $C_{ox}$  is the oxide capacitance ( $F/m^2$ )
- › W and L are the width and length of the channel respectively
- ›  $C_{ox}$  is given by

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}$$

- ›  $\epsilon_{ox}$  is permittivity and  $t_{ox}$  is oxide thickness ( $\epsilon_{ox} = 3.45 \times 10^{-11}$  F/m for Si)

## Applying a small $v_{DS}$

- › When a small voltage  $v_{DS}$  is applied a current  $i_D$  starts to flow through the induced n-channel
- › This current ( $i_D$ ) is carried by free electrons traveling from source to drain
- › This drain current is defined in terms of charge per unit length of n-channel and the electron drift velocity
- ›  $i_D = (\text{charge per unit length})(\text{electron drift velocity})$

$$i_D = (C_{ox} W v_{OV}) \left( \frac{\mu_n v_{DS}}{L} \right) = q t = q L \left( \frac{1}{L} \right) = q L (\text{vel})$$

$$i_D = \left[ (\mu_n C_{ox}) \left( \frac{W}{L} \right) v_{OV} \right] v_{DS} \quad \begin{matrix} \mu = \text{vel} / \vec{E} \\ \text{vel} = \mu * \vec{E} \end{matrix}$$

- › For small values of  $v_{DS}$ , the n-channel acts like a variable resistance whose value is controlled by  $v_{OV}$   $\vec{E} = \frac{V_{DS}}{L}$

## Applying a small $v_{DS}$

- › Conductance is defined as

$$g_{DS} = \frac{i_D}{v_{DS}} = (\mu_n C_{ox}) \left( \frac{W}{L} \right) v_{OV}$$

- › Conductance  $g_{DS}$  depends on:
  - $(\mu_n C_{ox})$  which is determined by the manufacturing process
  - $\left( \frac{W}{L} \right)$  or aspect ratio which depends on the geometry of the n-channel
  - $v_{OV} = v_{GS} - V_t$  which is applied externally

- › Process transconductance parameter is defined

$$k_n' = \mu_n C_{ox}$$

- › Multiplying by the aspect ratio yields the transconductance parameter

$$k_n = k_n' \frac{W}{L} = \mu_n C_{ox} \frac{W}{L}$$

## Applying a small $v_{DS}$

- For small values of  $v_{DS}$  the MOSFET behaves as a linear resistance  $r_{DS}$

$$r_{DS} = \frac{1}{g_{DS}} = \frac{v_{DS}}{i_D} = \frac{1}{(\mu_n C_{ox}) \left(\frac{W}{L}\right) v_{OV}}$$

- $r_{DS}$  depends on:
  - $(\mu_n C_{ox})$  which is determined by the manufacturing process
  - $\left(\frac{W}{L}\right)$  or aspect ratio which depends on the geometry of the n-channel
  - $v_{OV} = v_{GS} - V_t$  which is applied externally

## Operation as $v_{DS}$ is increased

- The charge in the tapered channel is proportional to the channel area which is proportional to

$$\frac{1}{2} [V_{OV} + (V_{OV} - v_{DS})] = V_{OV} - \frac{1}{2} v_{DS}$$

- From previous results

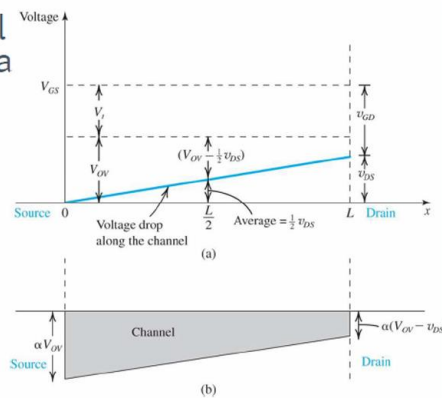
$$i_D = \left[ (\mu_n C_{ox}) \left(\frac{W}{L}\right) v_{OV} \right] v_{DS}$$

$$i_D = \left[ k'_n \left(\frac{W}{L}\right) v_{OV} \right] v_{DS}$$

- Substituting from above for  $V_{OV}$

$$i_D = \left[ k'_n \left(\frac{W}{L}\right) \left(V_{OV} - \frac{1}{2} v_{DS}\right) \right] v_{DS}$$

For a MOSFET with  $v_{GS} = V_t + v_{OV}$  application of  $v_{DS}$  causes the voltage drop along the channel to vary linearly, with an average value of  $v_{DS}$  at the midpoint. Since  $v_{GD} > V_t$ , the channel still exists at the drain end. (b) The Channel shape corresponding to the situation in (a) While the depth of the channel at the source is still Proportional to  $V_{OV}$ , the drain end is not.



## Operation as $v_{DS}$ is increased

$$i_D = \left[ k'_n \left(\frac{W}{L}\right) \left(V_{OV} - \frac{1}{2} v_{DS}\right) \right] v_{DS}$$

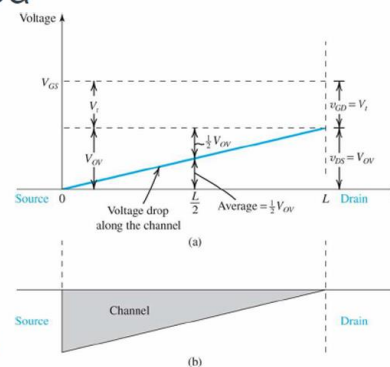
$$i_D = \left[ k'_n \left(\frac{W}{L}\right) \left(V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2\right) \right]$$

- As  $v_{DS}$  is increased to  $v_{OV}$ ,  $v_{GD}$  decreases to  $V_t$

- and the channel length at the drain reduces to zero

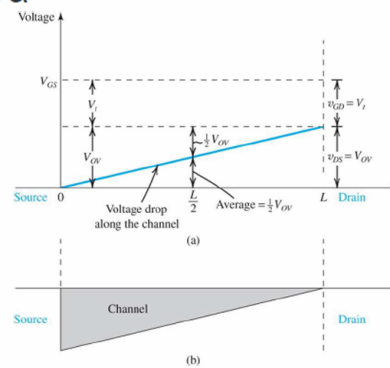
- At this point, the MOSFET enters saturation mode of operation

- Further increasing  $v_{DS}$  (beyond  $v_{OV}$ ) has no effect on the channel shape and  $i_D$  remains constant



## Operation as $v_{DS}$ is increased

- > The drain current saturates at  $v_{DS}=V_{OV}$
- >  $i_D = \left[ k'_n \left( \frac{W}{L} \right) \left( V_{OV} v_{DS} - \frac{1}{2} v_{DS}^2 \right) \right]$
- > Substituting  $v_{DS}=V_{OV}$  above
- >  $i_D = \left[ k'_n \left( \frac{W}{L} \right) \left( V_{OV} V_{OV} - \frac{1}{2} V_{OV}^2 \right) \right]$
- >  $i_D = \frac{1}{2} k'_n \left( \frac{W}{L} \right) V_{OV}^2$



## $i_D$ versus $v_{DS}$ for an NMOS

